AUTOMATED GAS LEAKAGE DETECTION SYSTEM

ABSTRACT

The **Automated Gas Leakage Detection System** is an advanced safety solution designed to detect gas leaks and automatically shut off the gas supply to prevent potential hazards. The system integrates an **MQ2 gas sensor** to continuously monitor levels in the air, an **Arduino Uno microcontroller** to process sensor data, and a **servo motor** to regulate the gas supply by automatically turning off the valve when a leak is detected.

Additionally, the system provides real-time **visual and audio alerts** using an LED and buzzer, ensuring immediate user awareness. Unlike conventional gas leakage detection systems that only provide alerts, this system takes proactive action by cutting off the gas supply, significantly reducing the risk of fire or explosion.

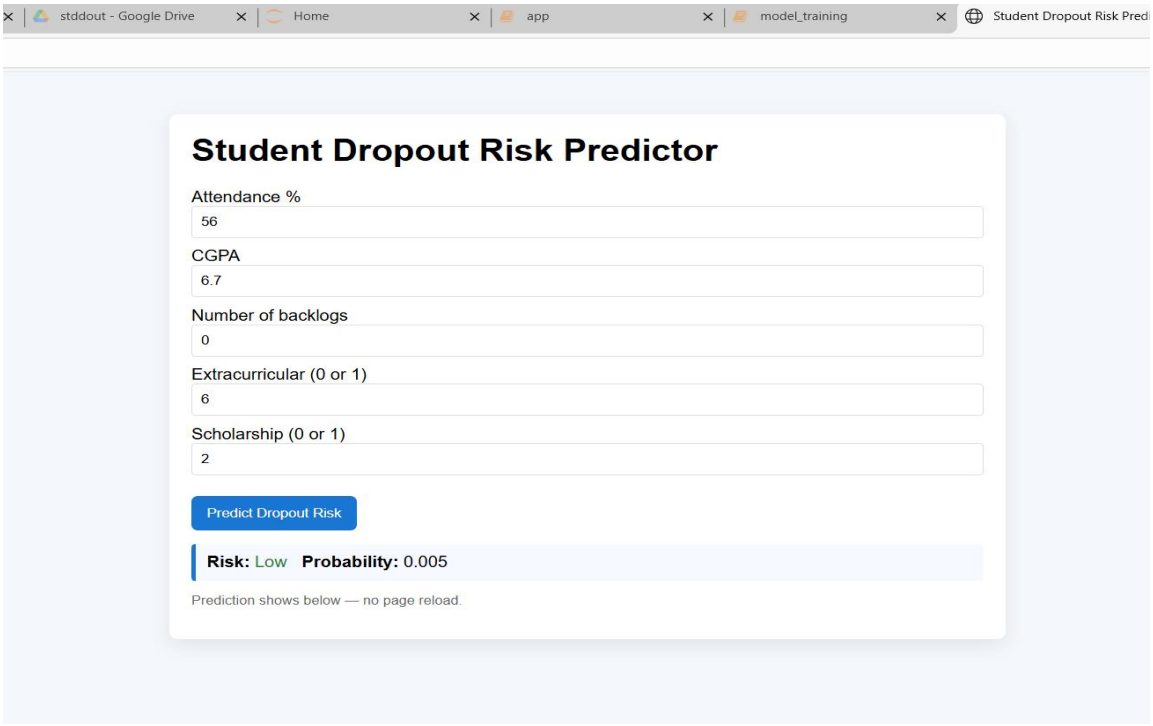
The project is designed for **household and industrial applications**, enhancing safety standards by eliminating the need for human intervention. The system is compact, cost-effective, and easy to implement, making it a reliable solution for preventing gas-related accidents.

STUDENT DROPOUT RISK PREDICTION USING MACHINE LEARNING

ABSTRACT

Student dropout is a growing concern in higher education. Institutions face significant challenges in identifying at-risk students before they leave. This project focuses on developing a **machine learning-based predictive system** that determines the likelihood of a student dropping out early. Using academic and engagement features such as attendance percentage, CGPA, number of backlogs, extracurricular participation, and scholarship status, the system classifies students into **Low, Medium, or High risk categories**. A **Random Forest Classifier** was implemented due to its robustness, accuracy, and ability to handle both categorical and numerical data. Exploratory Data Analysis (EDA) was performed to understand relationships between variables and dropout trends. The model achieved **92% accuracy**, showing strong predictive performance. Finally, a **Flask web application** was developed, allowing real-time prediction where users can input student details and instantly receive a dropout risk assessment. This system can assist educators in taking timely interventions, thereby improving student retention and institutional performance.

OUTPUT VIEW


stddout - Google Drive
Home
app
model_training
Student Dropout Risk Predi
Student Dropout Risk Predictor
Attendance %
56
CGPA
6.7
Number of backlogs
0
Extracurricular (0 or 1)
6
Scholarship (0 or 1)
2
Predict Dropout Risk
Risk: Low Probability: 0.005
Prediction shows below — no page reload.